

DURHAM, AustraliaWMO: **954870**Lat: **27.401S**Lon: **141.811E**Elev: **115**StdP: **99.95**Time Zone: **10.00 (E10)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	6.4	-12.0	1.4	26.1	-9.8	1.7	25.4	10.5	21.1	9.3	19.2	4.0	140	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	42.7	20.0	41.3	19.8	39.9	19.7	24.7	31.4	23.9	31.2	23.1	31.6	5.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.3	18.3	26.9	22.0	16.9	26.8	20.5	15.4	27.3	75.5	31.4	72.5	31.4	69.0	31.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.9	8.9	DB	2.9	44.7	1.2	1.6	2.0	45.9	1.3	46.8	0.6	47.7	-0.3	48.9	(4)
(5)			WB	0.6	25.5	1.1	1.0	-0.2	26.2	-0.8	26.8	-1.5	27.3	-2.2	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	32.3	30.7	28.1	23.7	18.2	14.4	14.1	16.1	20.3	24.6	27.6	30.5	(6)	
	DBStd	7.29	3.45	3.64	3.53	3.57	3.27	3.02	2.98	3.49	4.03	4.20	4.09	3.72	(7)	
	HDD10.0	2	0	0	0	0	0	1	1	0	0	0	0	0	(8)	
	HDD18.3	427	0	0	0	3	44	126	136	89	24	4	1	0	(9)	
	CDD10.0	4868	691	580	562	412	254	132	129	188	308	451	527	636	(10)	
	CDD18.3	2252	433	347	303	165	39	7	5	19	82	197	277	377	(11)	
	CDH23.3	31938	6945	5234	4062	1776	311	36	47	237	1015	2631	3861	5784	(12)	
(13)	CDH26.7	19111	4690	3337	2304	781	58	2	4	67	435	1404	2290	3738	(13)	
(14)	Wind	WSAvg	5.1	5.5	5.5	5.3	4.9	4.5	4.4	4.8	5.1	5.4	5.4	5.5	(14)	
(15) Precipitation	PrecAvg	212	36	32	22	14	14	11	12	7	12	9	24	17	(15)	
	PrecMax	703	135	210	123	79	94	52	88	32	63	60	114	100	(16)	
	PrecMin	69	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	132	38	44	30	23	22	14	23	9	19	12	25	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.5	43.8	41.3	36.0	30.3	26.5	27.2	32.0	36.6	39.1	43.1	43.8	(19)	
		MCWB	20.5	21.0	20.1	19.2	16.3	15.8	14.3	15.9	16.5	17.9	19.5	18.9	(20)	
	2%	DB	42.8	41.7	38.9	34.2	28.2	24.2	24.5	28.4	33.6	37.1	40.5	41.9	(21)	
		MCWB	20.2	20.1	19.4	18.3	15.9	14.9	13.5	13.9	15.4	17.0	18.7	19.0	(22)	
	5%	DB	41.3	40.0	37.0	32.7	26.7	22.1	22.6	25.8	31.3	35.6	38.3	40.1	(23)	
		MCWB	20.3	20.0	18.9	17.6	15.3	13.7	12.6	12.7	14.4	16.8	17.8	18.9	(24)	
	10%	DB	39.8	38.3	35.5	31.2	24.9	20.2	20.6	23.6	29.1	33.8	36.2	38.6	(25)	
		MCWB	20.4	20.0	18.9	16.8	14.5	12.9	11.9	12.0	13.9	16.3	17.6	19.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.8	25.5	24.6	22.7	19.7	18.0	18.9	17.2	19.5	20.8	22.6	24.4	(27)	
		MCDB	32.3	31.7	27.2	28.3	24.7	22.0	21.7	25.5	24.3	29.6	30.5	31.0	(28)	
	2%	WB	24.8	24.6	23.5	20.8	17.8	16.7	15.9	15.5	17.5	19.8	21.6	23.0	(29)	
		MCDB	31.1	32.8	28.4	29.5	24.3	21.3	20.4	24.6	26.9	29.9	30.4	32.1	(30)	
	5%	WB	23.9	23.9	22.6	19.6	16.5	15.3	14.1	13.9	16.3	18.6	20.8	21.9	(31)	
		MCDB	31.5	32.2	29.6	28.9	24.6	20.1	20.3	24.9	27.4	30.8	30.1	32.2	(32)	
	10%	WB	23.0	22.9	21.6	18.3	15.4	13.9	12.4	12.7	15.0	17.3	20.0	21.0	(33)	
		MCDB	32.9	31.2	30.0	28.2	23.3	19.0	19.5	22.5	27.8	31.2	30.4	33.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	11.7	11.7	12.0	11.6	10.8	11.7	12.9	13.4	13.9	13.1	12.6	(35)	
		MCDBR	13.1	13.2	13.2	12.9	12.8	12.5	13.7	15.2	16.2	15.3	15.3	14.0	(36)	
	5% WB	MCWBR	4.1	4.2	5.0	5.1	5.9	6.2	6.6	7.1	6.8	6.2	5.5	5.1	(37)	
		MCWBR	10.5	9.7	9.7	10.8	10.8	9.9	11.4	13.6	12.5	13.1	12.0	11.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.365	0.359	0.342	0.325	0.308	0.297	0.299	0.314	0.361	0.359	0.368	0.364	(40)	
		taud	2.440	2.481	2.505	2.513	2.510	2.535	2.511	2.445	2.321	2.348	2.379	2.411	(41)	
	Ebn at Noon		977	966	953	923	894	883	896	920	917	955	967	980	(42)	
		Edh at Noon	122	114	106	95	86	80	85	100	125	129	129	126	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.88	7.37	6.47	5.36	4.19	3.60	3.95	4.93	6.10	7.24	7.61	7.91	(44)	
	RadStd		0.32	0.48	0.41	0.29	0.19	0.21	0.20	0.24	0.37	0.38	0.46	0.43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+1.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page